MILITARY SPECIFICATION SHEET

INSTALLING AND REMOVAL TOOLS, CONNECTOR ELECTRICAL CONTACT, TYPE I AND II, CLASS 2, COMPOSITION A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the tools described herein shall consist of this document and the latest issue of specification MIL-I-81969.

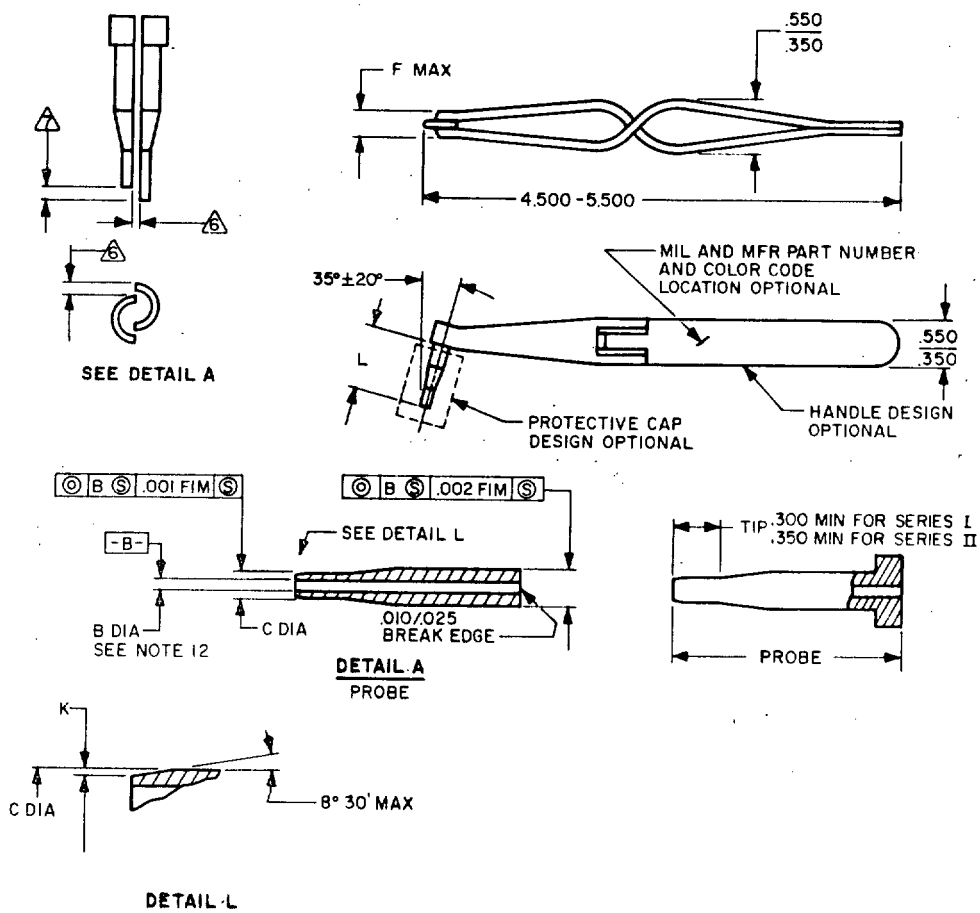


FIGURE 1. Installing tool (-01, -03, -05, -07, -09, -11, -13, -205, -207, -209).

1/ For complete supersession data, see page 6 of this specification.

AMSC N/A
FSC 5120
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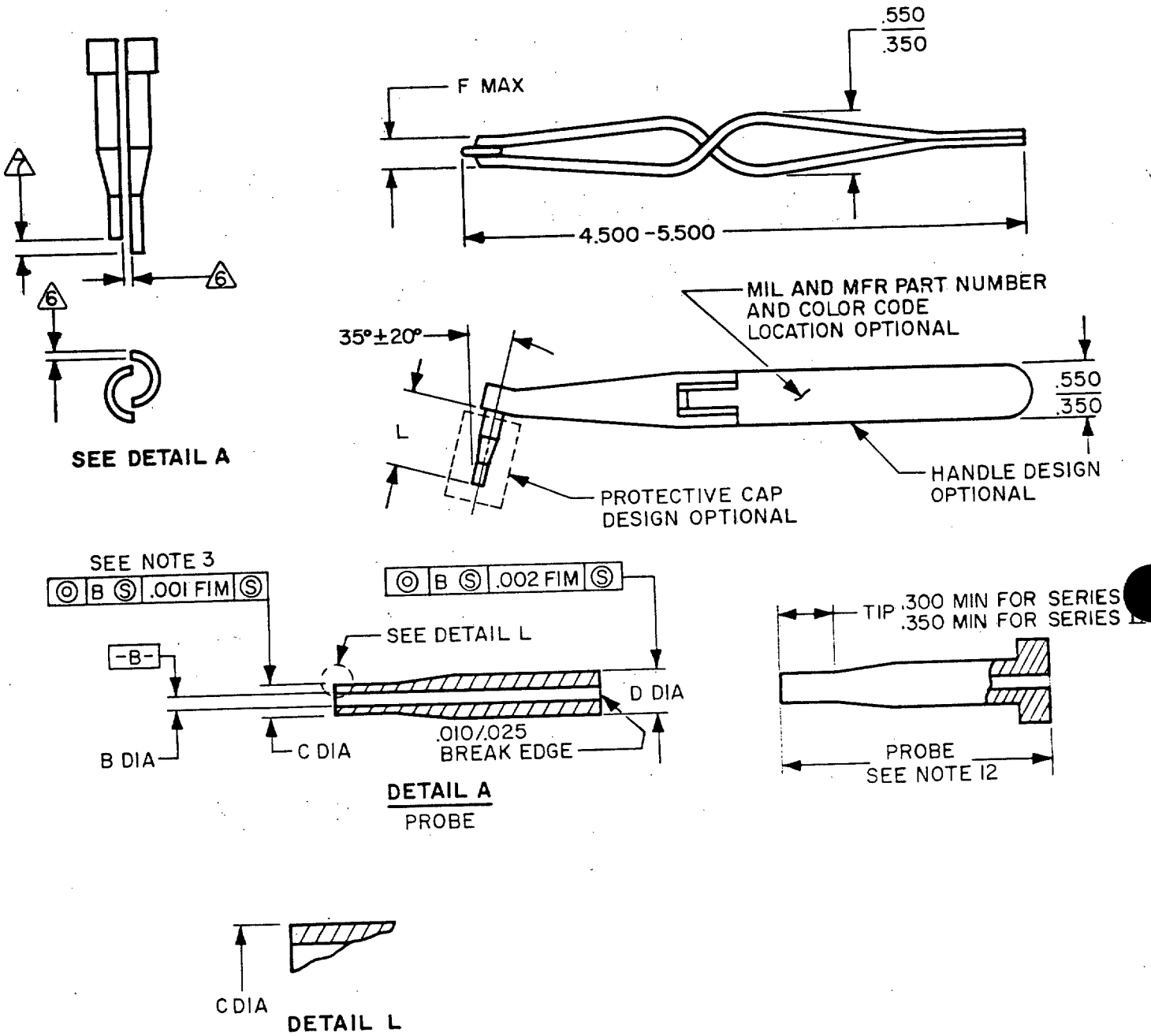


FIGURE 2. Removal tool (-02, -04, -06, -08, -10, -12, -14, -206, -208, -210).

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TABLE I. Dash numbers, dimensions.

Dash number	Size	Color code			A 1/ Dia +.0000 -.0002	B 2/ Dia +.001 -.001	C Dia +.000 -.001	D Dia ±.003	F Max	H 3/ Dia +.0000 -.0002	J 3/ Dia +.0002 -.0000	K	L ±.015
		No. 1	No. 2	No. 3									
-01	22D 22M	Green	---	---	.0545	.053	.0645	.113	.232	4/ .0645	.0515	.004	.650
-02		Green	White	---								.002	
-03	22	Brown	---	---	.0625	.061	.074	.113	.236	.0745	.0595	.0045	.650
-04		Brown	White	---								.0025	
-05	20	Red	---	---	.0855	.084	.097	.113	.254	.0975	.0825	.0045	.650
-06		Red	White	---								.0025	
-07	16	Blue	---	---	.1115	.110	.133	.135	.287	.1335	.1085	.007	.650
-08		Blue	White	---								.005	
-09	12	Yellow	---	---	.1595	.158	.187	.189	.335	.1875	.1565	.008	.650
-10		Yellow	White	---								.006	
-11	10	Green	---	---	.2165	.215	.249	.251	.401	.2495	.2135	.010	.650
-12		Green	White	---								.008	
-13	8	Red	---	---	.2795	.278	.317	.320	.475	.3175	.2765	.016	1.100
-14		Red	White	---								.012	
-205	20	Red	---	Black	.0855	.084	.104	.113	.262	.1045	.0825	.005	.950
-206		Red	White	Black								.003	
-207	16	Blue	---	Black	.1115	.110	.134	.137	.291	.1345	.1085	.007	1.200
-208		Blue	White	Black								.005	
-209	12	Yellow	---	Black	.1625	.161	.191	.194	.343	.1915	.1595	.008	1.200
-210		Yellow	White	Black								.006	

1/ See Note 6.
2/ See Note 12.
3/ See Notes 6 & 8.
4/ Tolerance +.0002
-.0000

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Inch	mm	Inch	mm	Inch	mm	Inch	mm
.0002	0.01	.0825	2.10	.1875	4.76	.287	7.29
.001	0.03	.084	2.13	.189	4.80	.291	7.39
.002	0.05	.085	2.16	.191	4.85	.300	7.62
.003	0.08	.0975	2.47	.1915	4.86	.317	8.05
.004	0.10	.1045	2.65	.194	4.93	.3175	8.06
.0045	0.11	.1085	2.76	.2135	5.42	.320	8.13
.007	0.18	.110	2.79	.215	5.46	.335	8.51
.008	0.20	.1110	2.85	.216	5.49	.343	8.71
.010	0.25	.113	2.87	.232	5.89	.350	8.89
.015	0.38	.1335	3.39	.236	5.99	.401	10.19
.025	0.64	.134	3.40	.2495	6.34	.475	12.07
.0515	1.31	.1345	3.41	.250	6.35	.550	13.97
.053	1.35	.137	3.48	.251	6.38	.650	16.51
.054	1.37	.1565	3.98	.254	6.45	.950	24.13
.0595	1.51	.158	4.01	.262	6.65	1.100	27.94
.061	1.55	.1590	4.04	.274	6.96	1.200	30.48
.062	1.57	.1595	4.05	.2765	7.02	4.500	114.30
.0645	1.64	.161	4.09	.278	7.06	5.500	139.70
.0740	1.88	.162	4.11				

NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.005 inch.
3. Dimensioning and tolerancing in accordance with ANSI Y14.5M.
4. Metric equivalents are given for general information only and are based upon 1.000 inch = 25.40 mm.
5. Concentricity to be maintained for a length of .250 inch minimum.
6. With probes closed around on "A" diameter pin, the probes shall aline themselves within .001 inch throughout the L length and clearance between probes shall be .000 to .006 inch for the installing tools and .000 to .002 for removal tools for the entire length of the probe. This clearance shall be central to the center line of the C diameter within ±.002 inches.
7. With the probes in the free state position, the probes shall end-aline within .005 inch.
8. The probes shall accept a ring gage of "H" diameter to a depth of .300 inch minumum for SI and .350 inch minimum for SII, with the probes in the ring gage a pin of "J" diameter shall enter the probes to a depth of .300 inch minimum for SI and .350 inch minimum for SII. As an option, the pin may be inserted prior to the ring gage. The A diameter pin shall not enter the tool.

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9. Tweezers shall be reverse action with box joint.

10. Surface on the exterior of the probes shall be 32 microinches maximum rms in accordance with ANSI B46.1.

11. Each tool shall be legibly and permanently marked on the handle in accordance with MIL-STD-130.

12. B diameter tolerance shall be +.001-.001 for tip, and +.010 -.001 for balance of probe.

REQUIREMENTS:

Dimensions and configuration.

Installing tool: See figure 1 and table I.
Removal tool: See figure 2 and table I.

Material: Handle - Corrosion resistant steel or QQ-S-700 steel treated to resist corrosion; 410, 416 stainless steel, 300 maraging steel, Almar 362 or approved suitable material.

Probes: Steel SAE 4130, AMS5643, AMS6370, 410, 416 stainless steel, 300 maraging steel, Croloy 16-6PH or approved suitable material, heat treated Rockwell Rc 35-45. For material too thin for accurate Rockwell measurement, the Knoop hardness scale shall be used. Knoop hardness (500 gram load minimum) shall be 350-460.

Plating: For non-corrosion resistant steels, electroless nickel in accordance with MIL-C-26074.

Plunger force: Not applicable.

Part number: Installing and removal tool: M81969/8 (and suitable dash number).

Application and Supersession Data: These tools shall be used to install and remove contacts from connectors as specified in table II.

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TABLE II

Military Part Number			
	For use with: Series I tools	For use with: Series II tools	Superseded Part Numbers
Size	MIL-C-38999 MIL-C-55302/68, /69, /166 thru /171 MIL-T-81714 (Size 22D) MIL-C-83733 (Size 22D)	MIL-C-5015 (Series MS 3450) MIL-C-26482 (Series 2) MIL-C-81703 (Series III) MIL-T-81714 (except 22D) MIL-C-83723 MIL-C-83733 (except 22D)	
22D	M81969/8-01	---	M55302/79-01 M27495A22M
	M81969/8-02	---	M55302/79-02 M27495R22M
22	M81969/8-03	---	M27495A22
	M81969/8-04	---	M27495R22
20	M81969/8-05	M81969/8-205	M27495A20
	M81969/8-06	M81969/8-206	M27495R20
16	M81969/8-07	M81969/8-207	M27495A16
	M81969/8-08	M81969/8-208	M27495R16
12	M81969/8-09	M81969/8-209	M27495A12
	M81969/8-10	M82969/8-210	M27495R12
10	M81969/8-11	---	---
	M81969/8-12	---	---
8	M81969/8-13	---	---
	M81969/8-14	---	---

The symbol --- indicates not applicable.

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Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - MI, AR, AT, GL
Navy - EC, SH, OS
Air Force - 11, 99
DLA - ES

User activities:
Army - ME, AV
Navy - MC, YD
Air Force - 19

Preparing activity:
Navy - AS
(Project 5120-B072)

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